

CAMAGUEY INTL, Cuba

WMO: 782550

Lat: 21.420N

Lon: 77.848W

Elev: 413

StdP: 14.48

Time Zone: -5.00 (CUB)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	59.0	62.0	48.8	51.8	68.7	51.6	57.5	70.2	24.9	77.1	21.6	78.6	5.1	10	0.352

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	16.5	92.9	74.5	91.6	74.8	90.7	74.8	79.2	85.8	78.5	85.2	77.9	84.8	8.6	70

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
77.4	144.7	81.0	77.0	142.6	80.7	75.7	136.8	80.3	42.9	86.0	42.1	85.8	41.4	84.8	84.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
21.1	18.8	16.8	DB	55.0	95.0	2.7	1.4	53.1	96.0	51.5	96.8	50.0	97.6	48.0	98.6	
			WB	51.4	80.2	3.0	1.3	49.2	81.1	47.4	81.9	45.7	82.6	43.5	83.5	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	78.3	73.2	74.5	75.6	78.0	79.6	81.3	82.4	82.4	81.6	79.9	76.1	74.8
	DBStd	4.19	3.85	3.53	3.51	2.55	2.26	1.95	1.56	1.63	1.64	2.05	2.99	3.40
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD65	6	3	1	0	0	0	0	0	0	0	0	0	2
	CDD50	10329	720	687	793	840	919	938	1004	1005	948	926	782	767
	CDD65	4860	257	268	329	390	454	488	539	540	498	461	332	304
	CDH74	38232	1382	1738	2343	2951	3661	4402	5224	5137	4369	3551	1876	1598
	CDH80	14095	338	523	814	1089	1446	1765	2210	2159	1742	1207	428	374
Wind	WSAvg	8.2	8.7	9.0	9.1	8.9	8.3	7.7	8.5	7.8	6.5	7.1	8.1	8.5
Precipitation	PrecAvg	53.9	1.3	1.1	2.1	3.5	8.3	7.8	5.6	6.4	8.3	6.4	2.7	1.5
	PrecMax	67.0	5.2	4.4	5.7	8.3	16.7	13.3	11.5	12.1	13.4	29.7	11.1	3.7
	PrecMin	42.1	0.0	0.0	0.0	0.7	1.7	3.3	1.9	1.1	4.4	1.1	0.0	0.1
	PrecStd	5.7	1.2	1.3	1.5	1.8	4.2	2.7	2.6	2.5	2.2	5.7	2.5	1.0
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	87.8	89.7	91.7	91.8	93.2	93.2	93.5	93.6	92.9	90.8	88.1	87.9
		MCWB	69.3	68.7	69.7	70.8	72.7	75.6	74.5	75.4	75.4	75.0	74.6	72.2
	2%	DB	85.8	87.6	89.4	89.8	91.1	91.4	91.8	91.8	91.3	89.4	86.1	86.0
		MCWB	70.1	69.0	69.4	71.0	73.1	75.8	75.1	75.7	75.7	75.4	74.0	72.5
	5%	DB	83.9	85.8	87.4	88.0	89.4	89.8	91.2	91.2	89.8	87.9	84.3	84.1
		MCWB	70.0	69.4	69.2	71.1	72.9	75.7	75.2	75.8	76.0	75.4	73.5	71.9
	10%	DB	82.0	83.8	84.6	86.2	87.6	88.4	89.7	89.6	88.2	86.3	82.5	82.2
		MCWB	69.8	69.3	69.1	71.1	72.7	75.7	75.7	76.2	76.0	75.5	72.6	71.6
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	74.0	73.8	74.1	77.0	78.0	80.4	79.3	79.8	79.7	79.1	77.7	76.2
		MCDB	80.6	81.1	81.4	83.8	84.5	87.7	86.1	87.3	86.7	84.5	82.2	81.7
	2%	WB	72.8	72.5	72.9	75.2	76.7	79.0	78.5	78.8	78.7	78.3	76.4	74.7
		MCDB	79.6	79.9	80.7	82.4	83.8	85.0	85.5	86.4	85.6	83.9	81.9	80.6
	5%	WB	71.9	71.5	72.1	74.1	75.8	78.0	77.7	78.0	77.8	77.4	75.3	73.9
		MCDB	78.7	79.0	79.9	81.6	83.0	84.2	85.2	85.7	85.1	83.3	80.9	79.7
	10%	WB	71.0	70.7	71.2	73.0	74.9	77.1	77.0	77.2	77.1	76.6	74.4	72.9
		MCDB	77.6	78.2	79.0	80.6	82.1	83.9	84.9	85.1	84.6	82.7	79.6	78.6
Mean Daily Temperature Range	5% DB	MDBR	16.1	17.8	18.6	17.8	16.9	16.0	16.5	16.6	15.8	14.0	13.4	14.5
		MCDBR	18.6	20.5	21.3	19.9	18.9	17.2	17.7	17.7	16.9	15.7	14.7	16.8
		MCWBR	6.5	6.3	6.1	5.5	5.7	6.1	5.7	6.1	6.0	5.9	5.9	6.0
	5% WB	MCDBR	16.4	17.8	18.4	17.3	16.3	15.9	16.3	16.3	15.5	14.1	12.7	14.7
MCWBR		6.5	6.3	6.3	6.2	6.2	6.8	6.3	6.5	6.4	6.2	6.0	5.9	
Clear-Sky Solar Irradiance	taub	0.370	0.377	0.394	0.417	0.450	0.495	0.491	0.470	0.427	0.411	0.394	0.426	
	taud	2.487	2.466	2.414	2.346	2.293	2.160	2.163	2.244	2.391	2.434	2.435	2.283	
	Ebn at Noon	286	291	289	282	270	256	257	264	275	276	275	261	
	Edh at Noon	32	34	38	41	43	49	48	45	38	35	33	38	
All-Sky Solar Radiation	RadAvg	1327	1601	1819	1992	1917	1909	2049	1977	1785	1535	1348	1248	
	RadStd	70	80	86	90	124	101	89	59	92	113	81	96	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (4 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page